

FigmaXR – Final Evaluation and Reflection (IP3)

Objective and Validation Metrics

The third prototype iteration of *FigmaXR* aimed to evaluate how transitioning from virtual reality (VR) to mixed reality (MR) affects the quality and authenticity of interface testing experiences. Specifically, this test compared two conditions:

- **Condition A (VR):** Controller-based interaction in a fully virtual space.
- **Condition B (MR):** Hand-tracking interaction using passthrough mode, integrating digital devices into the real physical environment.

The main objectives were to examine differences in (1) perceived naturalness, (2) physical comfort, and (3) contextual realism between the two modes. Success was defined as the majority of participants perceiving MR as more natural and realistic, with minimal fatigue or technical confusion during hand-based interaction.

Results

Six participants experienced both conditions in a counterbalanced order. Each completed identical tasks—grabbing, holding, and placing a virtual device—followed by a short interview.

Dimension	Key Observations	Supporting Evidence
Perceived Naturalness	5/6 participants described MR hand interaction as “more natural” and “closer to reality.”	Verbal feedback and observation notes
Physical Comfort	4 participants found MR more comfortable and less mentally taxing; 2 mentioned brief fatigue during extended gesture use.	Observation notes
Contextual Realism	All participants reported that seeing their real environment through passthrough mode made the interaction feel “more believable” and “connected to real work settings.”	Post-test interviews

Notable participant comments include:

“The MR mode feels like I’m testing on my desk rather than in a game.”

“Controllers are accurate but feel detached from the real world.”

“Hand-tracking sometimes missed, but it’s more fun and immersive.”

Overall, 5 of 6 participants preferred the MR version, citing its intuitive gestures and spatial awareness, though one user noted that controller-based VR still offered higher precision for fine selections.

Analysis / Insights

1. MR Enhances Realism and Embodied Presence

Participants consistently described the MR environment as more believable and spatially coherent, aligning with Billingham et al. (2015), who emphasize ecological validity as a defining strength of MR. By allowing users to see and move within their real physical workspace, *FigmaXR*’s MR mode bridged the gap between virtual prototype testing and authentic design contexts.

2. Hand-Tracking Feels Natural but Technically Imperfect

Echoing Pangestu et al. (2022), hand gestures were perceived as more intuitive but less precise than controller input. Small misregistrations occasionally broke the flow of interaction, suggesting that while MR increases realism, it may also introduce usability challenges that impact precision-critical tasks.

3. Context Awareness Supports Focus and Comfort

Several participants reported feeling calmer and more focused when testing in a familiar environment. The passthrough background reduced the sense of detachment associated with VR, minimizing cognitive dissonance between body movement and visual perception.

These patterns suggest that *FigmaXR* successfully increased interaction naturalness and presence but still requires refinement of gesture detection fidelity and user feedback mechanisms.

In summary, the testing outcomes confirmed the hypothesis that MR interaction fosters a more embodied and realistic interface evaluation experience than VR. However, gesture precision and visual feedback require further optimization to fully support professional-level design testing.

Reflection

Prototype Session Review

This testing session revealed that participants strongly appreciated the MR experience for its realism and embodied interaction. However, from a designer's perspective, the process also exposed significant practical challenges. While the concept of *FigmaXR* was ambitious and conceptually well-developed, the actual implementation often felt uncertain — I frequently found myself unsure about what exactly to build next or how to realize the intended level of interactivity. As a result, the prototype's interaction scope remained limited compared to the richness of the original vision. Technical constraints, especially within Unity's MR framework, made it difficult to implement advanced gesture-based interactions or precise object behaviors. These gaps reduced the overall sense of responsiveness and flow during testing.

Methodological Reflection

The A/B comparative design effectively isolated the difference between controller-based and hand-tracking interactions, yielding meaningful insights about usability and realism. However, the small participant pool restricted quantitative generalization. Furthermore, because the prototype itself had constrained interactivity, some participants struggled to fully explore the intended experience. This indicates that a more refined prototype—both technically and interactively—would be necessary for future testing to generate deeper feedback.

Concept Evaluation

The shift from VR to MR largely validated the theoretical premise that embodied interaction and spatial context improve the quality of interface testing. Yet, this round also exposed the gap between design intention and technical execution. While the idea of a seamless MR prototyping tool is strong, my limited technical proficiency in Unity's mixed reality development prevented the concept from being fully realized. This experience emphasized the importance of balancing creative ambition with technical feasibility and iterative implementation.

Improvements and Extensions

Moving forward, the most immediate improvement lies in expanding my technical fluency in Unity's XR Interaction Toolkit and MR-specific scripting. This would allow me to implement more dynamic gestures, haptic responses, and contextual feedback loops that make the concept feel genuinely interactive rather than conceptual. Additionally, collaborating with technically specialized peers or exploring plugin-based MR frameworks could accelerate the prototype's functionality. Conceptually, *FigmaXR* could evolve into a hybrid MR/AR evaluation system—bridging the accessibility of AR with the immersion of MR—to support realistic yet achievable design testing workflows.

Conclusion

The IP3 evaluation of *FigmaXR* demonstrated a clear conceptual success but also exposed practical limitations. The transition from VR to MR effectively enhanced realism, user engagement, and ecological validity, confirming the value of testing digital interfaces within real physical contexts. Most participants preferred the MR version, describing it as more intuitive, immersive, and natural.

However, the project also revealed the gap between design ambition and technical feasibility. While the conceptual vision of *FigmaXR* was rich—imagining seamless mixed-reality testing through intuitive gestures—the actual development process proved much more challenging.

Implementing stable MR interactions in Unity required a level of technical fluency that I have not yet achieved. As a result, the prototype's interactivity remained limited, and certain design ideas could not be fully realized. At times, this made the building process feel uncertain and directionless, as I struggled to translate conceptual ideas into functioning features. This experience highlighted an important personal and professional insight: technical skill is as essential as conceptual creativity in XR design. To move from vision to execution, I must strengthen my Unity programming capabilities and develop a more systematic prototyping workflow. With improved technical grounding, *FigmaXR* could evolve from a promising concept into a robust, deployable tool for immersive UI evaluation—bridging imagination and implementation in extended reality.

References

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